Predictive Current Control of a Voltage Source Inverter

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***Abstract*—This paper presents a predictive current control method and its application to a voltage source inverter. The method uses a discrete-time model of the system to predict the future value of the load current for all possible voltage vectors generated by the inverter. The voltage vector which minimizes a quality function is selected. The quality function used in this work evaluates the current error at the next sampling time. The performance of the proposed predictive control method is compared with hysteresis and pulsewidth modulation control. The results show that the predictive method controls very effectively the load current and performs very well compared with the classical solutions.**

***Index Terms*—Current control, digital control, inverters, predictive control.**

I. Introduction

CURRENT control of a three-phase inverter is one of the most important and classical subjects in power electronics and has been extensively studied in the last decades. Nonlinear methods, like hysteresis control and linear methods, like proportional-integral controllers using pulsewidth modulation (PWM) are well documented in literature [1]–[3].

With the development of powerful and fast microprocessors, increasing attention has been dedicated to predictive current control. In this method, load and converter models are used to predict current behavior, and thereby select the most appropriate actuation following an arbitrary control criteria [4]–[11]. Predictive control is a very wide concept and different control methods have been presented under this name. A classification of them is presented in [4].

One approach uses predictive control to calculate the necessary load voltage to optimize the current behavior. Later, a modulator is used to generate this desired voltage. In this approach, the converter is simply modeled as a gain. This strategy has been used in current control for inverters [6], [7], as well as for rectifiers and active filters [8]. A variation to this method calculates the duty cycle of the PWM pulses necessary for the current control [9], [10].

One advantage of predictive control is the possibility to include nonlinearities of the system in the predictive model, and hence calculate the behavior of the variables for different conduction states. This property was exploited in an earlier study [12], where predictive control was used to minimize switching frequency for high-power inverters. Also in [11], this property of predictive control is used to evaluate the behavior of the current error for each switching state in a single-phase active filter.

A conceptually different approach is presented in [13], to control a matrix converter. The model of the system is used to predict the behavior of the load and input current for each different switching state of the matrix converter. The switching state that minimizes a quality function is selected. This method demonstrates that the use of predictive control can avoid the use of complex modulation techniques.

This paper presents the method introduced in [13] and applied to a three-phase inverter. A detailed explanation of the method is presented, including the models used for current prediction and the quality function used for switching state selection. Simulation results comparing the performance of the proposed strategy with well-known hysteresis and PWM control, are shown. Finally, experimental results are presented to validate the theoretical studies.

II. Classical Control Methods

A. Hysteresis Current Control

In this control strategy, shown in Fig. 1, measured load currents are compared with the references using hysteresis comparators. Each comparator determines the switching state of the corresponding inverter leg (S_a, S_b, and S_c) such that the load currents are forced to remain within the hysteresis band.

This method is conceptually simple and the implementation does not require complex circuits or processors. The performance of the hysteresis controller is good, with a fast dynamic response. Due to the interaction between the phases, the current error is not strictly limited to the value of the hysteresis band.

The switching frequency changes according to variations of the load parameters and operating conditions. This is one of the major drawbacks of hysteresis control, since variable switching frequency can cause resonance problems. In addition, the switching losses restrict the application of hysteresis control to lower power levels.

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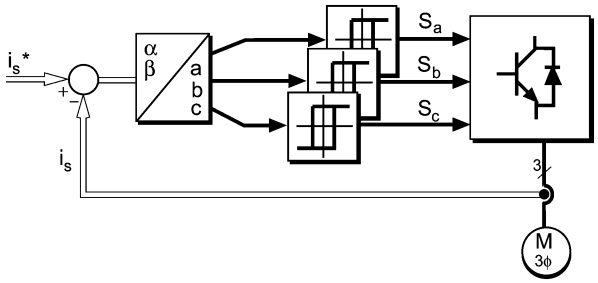


Fig. 1. Hysteresis current control.

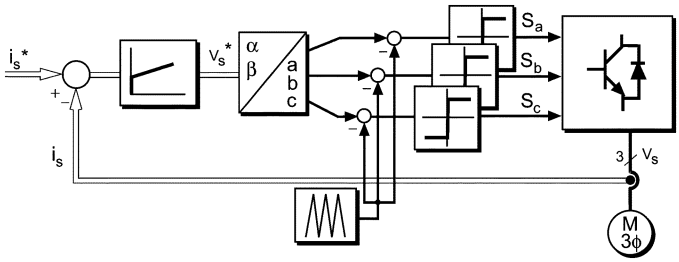


Fig. 2. PWM current control.

B. Linear Current Control With PWM

The PWM current control scheme is shown in Fig. 2. Here, the error between the reference and the measured load current is processed by a proportional-integral (PI) controller to generate the reference load voltages. A modulator is needed to generate the drive signals for the inverter switches. The reference load voltages are compared with a triangular carrier signal, and the output of each comparator is used to drive an inverter leg.

With this method, constant switching frequency, fixed by the carrier is obtained. The performance of this control scheme depends on the design of the controller parameters, and on the frequency of the reference current. Although the PI controller assures zero steady-state error for continuous reference, it can present such an error for sinusoidal references. This error increases with the frequency of the reference current and may become unacceptable for certain applications.

III. Description of Predictive Current Control

A. The Control Strategy

The proposed predictive control strategy is based on the fact that only a finite number of possible switching states can be generated by a static power converter and that models of the system can be used to predict the behavior of the variables for each switching state. For the selection of the appropriate switching state to be applied, a selection criteria must be defined. This selection criteria is expressed as a quality function that will be evaluated for the predicted values of the variables to be controlled. Prediction of the future value of these variables is calculated for each possible switching state. The switching state that minimizes the quality function is selected.

This control strategy can be summarized in the following steps.

- Define a quality function g.
- Build a model of the converter and its possible switching states.
- Build a model of the load for prediction.

A discrete-time model of the load is needed to predict the behavior of the variables evaluated by the quality function, i.e., the load currents.

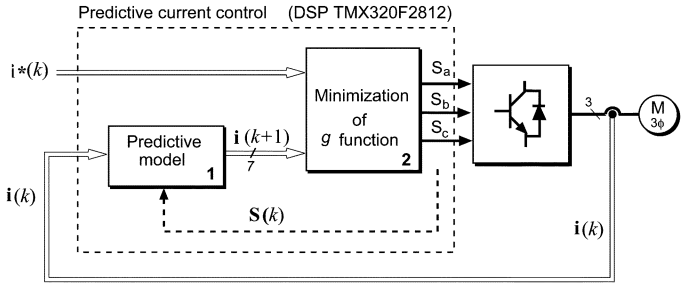


Fig. 3. Predictive current control block diagram.

A block diagram of the predictive control strategy applied to the current control for a three-phase inverter is shown in Fig. 3. The current control is performed in the following steps.

1) The value of the reference current $\mathbf{i}^*(k)$ is obtained (from an outer control loop), and the load current $\mathbf{i}(k)$ is measured.
2) The model of the system (block 1) is used to predict the value of the load current in the next sampling interval $\mathbf{i}(k+1)$ for each of the different voltage vectors.
3) In this case, the quality function g evaluates the error between reference and predicted currents in the next sampling interval. The voltage that minimizes the current error is selected and applied to the load (block 2).

B. Quality Function

The current error for the next sampling instant can be expressed in orthogonal coordinates as follows:

$$g = |i_\alpha^* - i_\alpha^p| + \left|i_\beta^* - i_\beta^p\right| \tag{1}$$

where i_α^p and i_β^p are the real and imaginary part of the predicted load current vector $\mathbf{i}(k+1)$, i_α^* and i_β^* are the real and imaginary part of the reference current.

Different control criteria will be expressed in different quality functions. In this work, the absolute error is used for computational simplicity. Other quality functions could evaluate the error integral over a sampling period, or the square error, for example. In [14], the torque and flux are directly controlled by evaluating the torque and flux error in the quality function. In the same way, active and reactive power are directly controlled in [15] for an AC/DC/AC converter. Also, additional terms can be added to the quality function to improve other aspects of the control like minimizing the switching frequency and DC link voltage balancing, as presented in [16], for a three-phase neutral point clamped inverter.

C. Inverter Model

The power circuit of the converter considered in this work is shown in Fig. 4.

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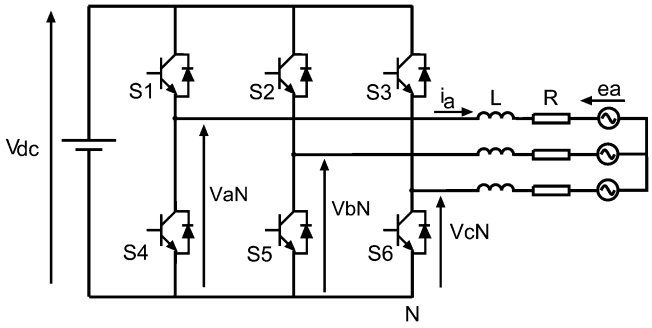


Fig. 4. Voltage source inverter power circuit.

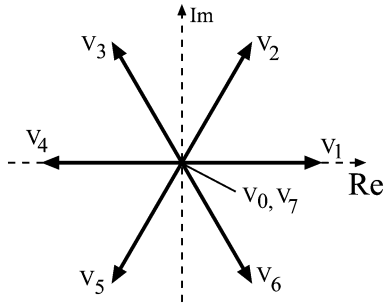


Fig. 5. Voltage vectors generated by the inverter.

The switching states of the converter are determined by the gating signals S_a, S_b, and S_c as follows:

$$S_a = \begin{cases} 1, & \text{if } S_1 \text{ on and } S_4 \text{ off} \\ 0, & \text{if } S_1 \text{ off and } S_4 \text{ on} \end{cases} \tag{2}$$

$$S_b = \begin{cases} 1, & \text{if } S_2 \text{ on and } S_5 \text{ off} \\ 0, & \text{if } S_2 \text{ off and } S_5 \text{ on} \end{cases} \tag{3}$$

$$S_c = \begin{cases} 1, & \text{if } S_3 \text{ on and } S_6 \text{ off} \\ 0, & \text{if } S_3 \text{ off and } S_6 \text{ on} \end{cases} \tag{4}$$

and can be expressed in vectorial form by

$$\mathbf{S} = \frac{2}{3}\left(S_a + \mathbf{a}S_b + \mathbf{a}^2 S_c\right) \tag{5}$$

where $\mathbf{a} = e^{j2\pi/3}$.

The output voltage space vectors generated by the inverter are defined by

$$\mathbf{v} = \frac{2}{3}\left(v_{aN} + \mathbf{a}v_{bN} + \mathbf{a}^2 v_{cN}\right) \tag{6}$$

where v_{aN}, v_{bN}, and v_{cN} are the phase to neutral (N) voltages of the inverter (Fig. 4). Then, the load voltage vector $\mathbf{v}$ can be related to the switching state vector $\mathbf{S}$ by

$$\mathbf{v} = V_{dc}\mathbf{S} \tag{7}$$

where V_{dc} is the DC link voltage.

Considering all the possible combinations of the gating signals S_a, S_b, and S_c, eight switching states, and consequently, eight voltage vectors are obtained. Note that $\mathbf{v}_0 = \mathbf{v}_7$, resulting in only seven different voltage vectors, as shown in Fig. 5.

Using modulation techniques like PWM, the inverter can be modeled as a linear system. Nevertheless, in this paper, the inverter is considered as a nonlinear discrete system with only seven different states as possible outputs.

A more accurate model of the converter model could be used for higher switching frequencies. It may include deadtime, insulated gate bipolar transistor (IGBT) saturation voltage, and diode forward voltage drop, for example. In this work, emphasis has been put in simplicity, so a simple model of the inverter will be used.

D. Load Model

In a balanced three-phase load, the current can be defined as a space vector by

$$\mathbf{i} = \frac{2}{3}\left(i_a + \mathbf{a}i_b + \mathbf{a}^2 i_c\right) \tag{8}$$

and the load EMF as

$$\mathbf{e} = \frac{2}{3}\left(e_a + \mathbf{a}e_b + \mathbf{a}^2 e_c\right). \tag{9}$$

In this way, the load current dynamics can be described by the vector equation

$$\mathbf{v} = R\mathbf{i} + L\frac{d\mathbf{i}}{dt} + \mathbf{e} \tag{10}$$

where R is the load resistance, L the load inductance, $\mathbf{v}$ the voltage generated by the inverter, and $\mathbf{e}$ the load back-EMF.

For simulation and experimental results, the load back-EMF is assumed to be a sinusoidal with constant amplitude and constant frequency.

E. Discrete-Time Model

A discrete-time form of the load current (10) for a sampling time T_s can be used to predict the future value of load current with the voltage and measured current at the kth sampling instant.

Approximating the derivative $d\mathbf{i}/dt$ by

$$\frac{d\mathbf{i}}{dt} \approx \frac{\mathbf{i}(k) - \mathbf{i}(k-1)}{T_s} \tag{11}$$

and replacing it in (10), the following expression is obtained for the future load current:

$$\mathbf{i}(k) = \frac{1}{RT_s + L}\left[L\mathbf{i}(k-1) + T_s\mathbf{v}(k) - T_s\mathbf{e}(k)\right] \tag{12}$$

where the term RT_s could be neglected if the sampling period is small enough and the load is mainly inductive.

Shifting the discrete-time one step forward in (12), the future load current can be determined by

$$\mathbf{i}(k+1) = \frac{1}{RT_s + L}\left[L\mathbf{i}(k) + T_s\mathbf{v}(k+1) - T_s\mathbf{e}(k+1)\right]. \tag{13}$$

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The load back-EMF can be estimated using (12) and measurements of the load voltage and current, resulting in the following expression:

$$\hat{\mathbf{e}}(k) = \mathbf{v}(k) + \frac{L}{T_s}\mathbf{i}(k-1) - \frac{RT_s + L}{T_s}\mathbf{i}(k) \qquad (14)$$

where $\hat{\mathbf{e}}(k)$ is the estimated value of $\mathbf{e}(k)$. The future back-EMF can be calculated using an extrapolation of the present and past values of the estimated back-EMF, or it can be supposed that the back-EMF does not change considerably in one sampling interval and, in that case, assume $\mathbf{e}(k+1) = \hat{\mathbf{e}}(k)$.

F. Voltage Vector Selection

In the proposed predictive algorithm, (13) is evaluated for each of the possible seven voltage vectors, giving seven different current predictions. The voltage vector whose current prediction is closest to the expected current reference is applied to the load at the next sampling instant. In other words, the selected vector will be the one that minimizes the quality function

$$g = \left|i_\alpha^*(k+1) - i_\alpha(k+1)\right| + \left|i_\beta^*(k+1) - i_\beta(k+1)\right|. \qquad (15)$$

However, the future reference current value $\mathbf{i}^*(k+1)$ required by (1) is unknown. Therefore, it has to be predicted from the present and previous values of the current reference using a second-order extrapolation given by

$$\mathbf{i}^*(k+1) = 3\mathbf{i}^*(k) - 3\mathbf{i}^*(k-1) + \mathbf{i}^*(k-2) \qquad (16)$$

obtained from the Lagrange extrapolation formula for $n = 2$ (quadratic) and is appropriate for a wide frequency range of $\mathbf{i}^*$ [7]. A similar extrapolation formula can be used to estimate $\hat{\mathbf{e}}(k+1)$.

For sufficiently small sampling times T_s, it can be assumed that $\mathbf{i}^*(k+1) \approx \mathbf{i}^*(k)$ and no extrapolation is needed. This approximation is considered in Fig. 3.

IV. Implementation of the Control Strategy

A. Practical Considerations

The control strategy has been implemented on a digital signal processor (DSP). The timing of the different tasks performed by the DSP is shown in Fig. 7. The time elapsed between the beginning of the sampling interval and the end of task 4 is about 7 μs.

It can be observed in Fig. 7 that the values for switching state to be applied in the time interval $k+1$ are calculated in the interval k. This is done in order to deal with the processing time delay, which is the most important delay on the system, fixing it to one sampling time. This delay has been included in the design of the control law for the experimental results, as well as for the simulations. Delays associated with the response of the gate drive circuitry and switching of the devices can be neglected, due to their small magnitude, even for high sampling rates.

Six digital outputs of the DSP are used to deliver the gate drive signals for the IGBTs. These outputs are set directly by the control algorithm and no modulator is needed.

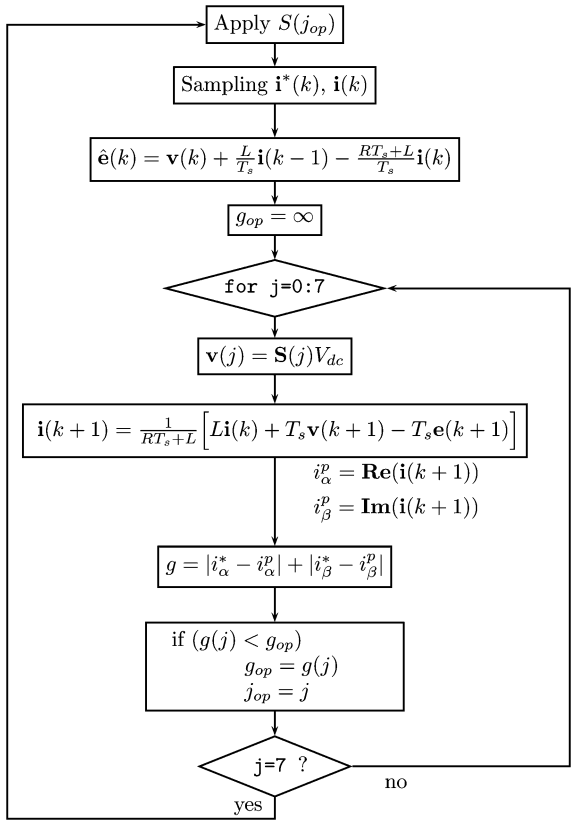


Fig. 6. Flow diagram of the implemented control algorithm.

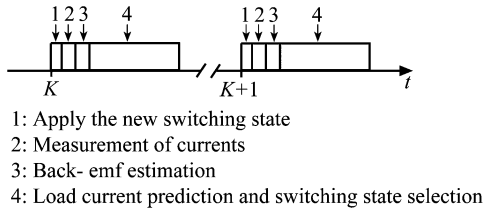


Fig. 7. Timing of the different tasks.

Two analog inputs of the DSP are needed for the measurement of the load currents. Two phases of the load current are measured and are used to calculate the current vector in orthogonal coordinates. The reference current is obtained from an outer control loop (e.g., speed control loop).

A table with all the possible switching states is used to generate the output signals to drive the IGBTs in the inverter. A corresponding table with the possible voltage vectors is used to calculate the prediction of the future load currents.

The future load currents are predicted for each voltage vector. The quality function is evaluated for each prediction. The index of the voltage vector that minimizes the quality function is stored. At the beginning of the next sampling period, the index value is used to read the table of switching states and generate the corresponding gate signals for the IGBTs.

B. Control Algorithm

The control algorithm is detailed in Fig. 6 as a flow diagram. As shown in the diagram, the minimization of the quality function can be implemented as a `for` cycle predicting for

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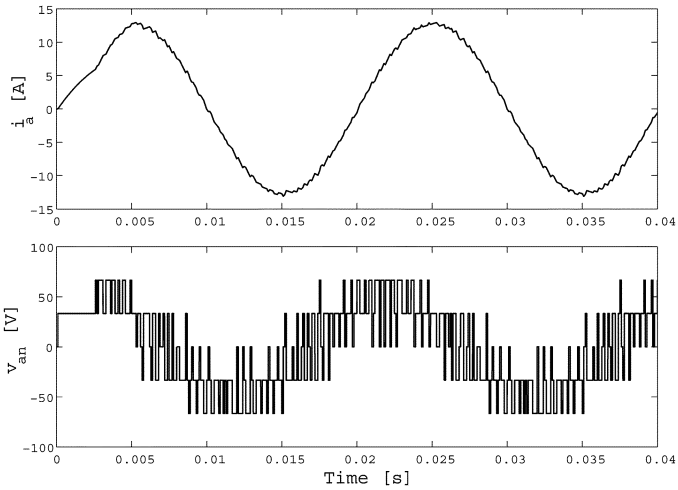


Fig. 8. Load current and load voltage for predictive current control.

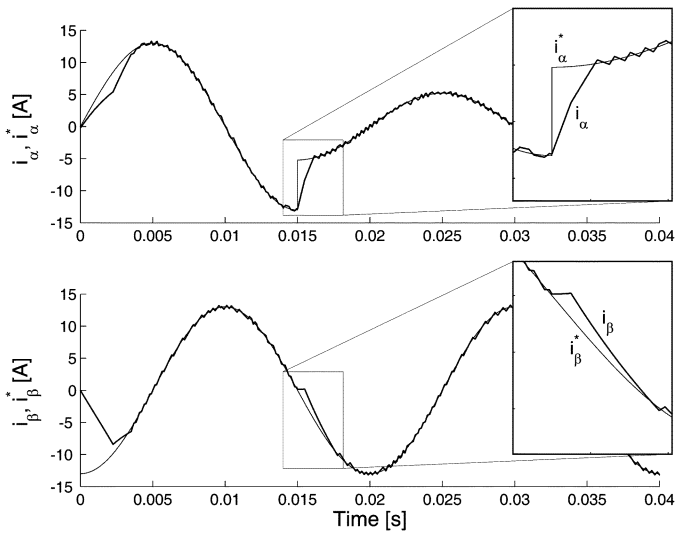


Fig. 9. Simulation results for a step in the reference current i_α^* for hysteresis current control.

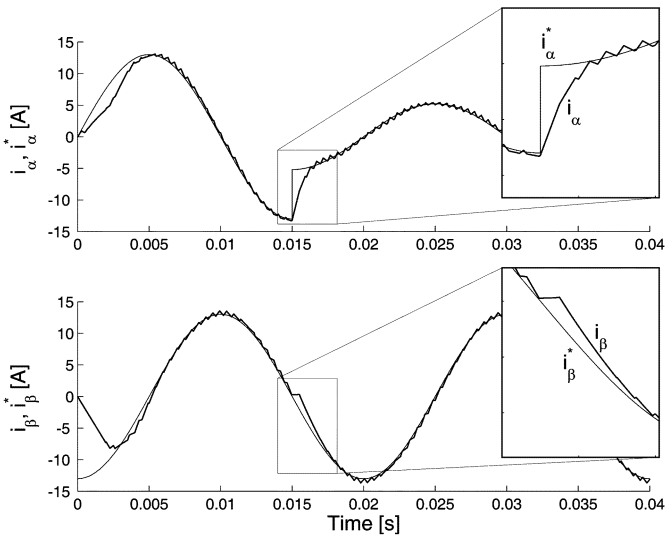


Fig. 10. Simulation results for a step in the reference current i_α^* for PWM current control.

each voltage vector, evaluating the quality function, and storing the minimum value and the index value of the corresponding switching state.

The control algorithm is implemented in a very simple way with the following program lines:

```
function S_min = conmuta(iref, ik)
%back − emf estimation
e1 = v_1 + (L/Ts) ∗ ik_1 − (R ∗ Ts + L)/Ts ∗ ik;
g_min = inf; for i = 1 : 7
%load current prediction
ik1 = [L ∗ ik + (Ts) ∗ (v(i) − e1)]/(R ∗ Ts + L);
g = abs(real (irefk1) − real (ik1))
+abs(imag(irefk1) − imag(ik1));
%minimization of g
if (g < g_min)
x_min = i;
g_min = g;
end
end
v_1 = v(x_min);
ik_1 = ik;
S_min = S(:, x_min);
```

V. Simulation Results

Simulations of an inverter controlled by the three different current control methods have been carried out with Matlab/Simulink, in order to assess the performance of the proposed predictive method, compared with the classical schemes.

The parameters of the simulated system are: $V_{dc} = 100$ V, $R = 0.5\ \Omega$, $L = 10$ mH, and the back-EMF is sinusoidal with fixed amplitude and frequency.

Fig. 8 shows the load current and voltage for the proposed predictive current control. It can be observed that the waveform of load voltage is very similar to a voltage generated with classical modulation techniques. In the first part, this result presents the transient behavior of the control system, starting from an initial current equal to zero. This result has been obtained with a sampling time of $T_s = 100\ \mu$s.

For comparison purposes, controller parameters of the classical methods considered in this work are designed to obtain comparable average switching frequencies. Namely, a hysteresis width of $\delta = \pm 0.3$ A and a PWM carrier frequency of 2 KHz.

A comparison of the proposed predictive current control with the conventional hysteresis and PWM control is presented in Figs. 9–11. Here, the amplitude of reference current i_α^* is reduced from 13 A to 5.2 A at instant $t = 0.015$ s, while keeping

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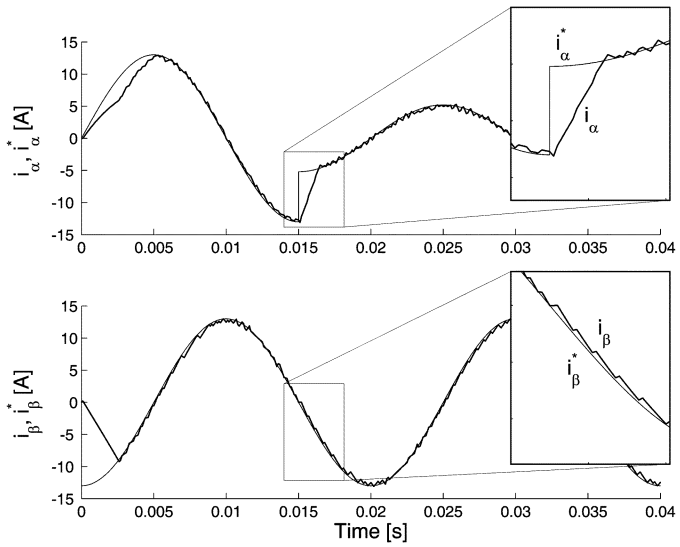


Fig. 11. Simulation results for a step in the reference current i_α^* for predictive current control.

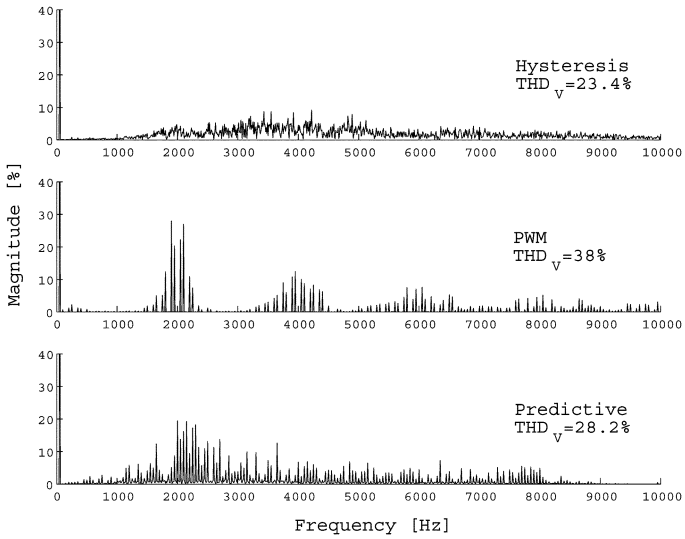


Fig. 12. Load voltage spectrum.

the amplitude of current i_β^* fixed. This is done to assess the decoupling capability of the current control loop. Hysteresis control, shown in Fig. 9, presents good dynamic response but with some noticeable coupling effects between i_α and i_β. The regulated currents using PWM modulation, shown in Fig. 10, present similar coupling behavior between i_α and i_β, and a slower response due to the dynamics of the closed current loops. The response of the proposed predictive current control, for the same test, is shown in Fig. 11. Its dynamic response is as fast as the one obtained by hysteresis control but with an inherent decoupling between both current components.

Besides the reference tracking capabilities of any current control method, another important performance measure is the output voltage spectrum generated by the inverter. The voltage spectrum for the three control methods are compared in Fig. 12.

In Fig. 12(a), it can be observed that hysteresis control produces a continuous and wide frequency range output voltage

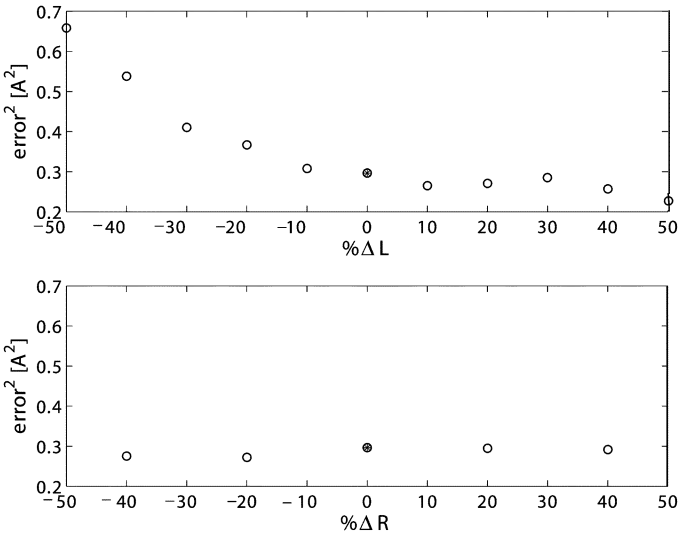


Fig. 13. Effect of model errors in the square error of the load current.

spectrum, which is considered a disadvantage of this method. The frequency spectrum of Fig. 12(b) shows that the harmonic content generated when using PWM current control, is concentrated around the carrier frequency. This is considered an advantage of PWM over hysteresis control. Finally, Fig. 12(c) presents the frequency spectrum obtained with predictive current control. The voltage spectrum of the proposed method is characterized by discrete spectral lines similar to those of PWM current control, although these spectral lines are more spread over the frequency range. A possible explanation for this, is the fact that the switching state of the inverter can be changed only once during each sampling instant, thus switching frequency is limited to 1/2 of the sampling frequency (f_s). However, switching states do not change in every sampling instant, therefore the average switching frequency is always less than $f_s/2$. Results show that the average switching frequency concentrates between $f_s/5$ and $f_s/4$.

A. *Effect of Load Model Errors*

Considering that the quality of the control strategy depends on the model used to predict the behavior of the load currents, the effect of load model errors has been studied by simulations.

The effect of errors in the value of the load inductance and resistance in the average square error of the load current is shown in Fig. 13. The load resistance R has a very small effect over the prediction and, in fact, it can be neglected. However, errors in the load inductance L have a major importance for the load current prediction, and hence, in the behavior of the current control. As shown in Fig. 13, estimating a lower value of the inductance has a deepest effect in the current error than estimating a higher value. Using a smaller value of L can increase the delay in the reference tracking, as shown in Fig. 14(a). For a higher value of L, the behavior of the load currents is shown in Fig. 14(b).

VI. Experimental Results

An experimental setup was developed using a DSP model TMS320F2812 for a sampling time $T_s = 100\ \mu$s and $T_s = 20\ \mu$s, and tested with an active load with values $R = 10\ \Omega$

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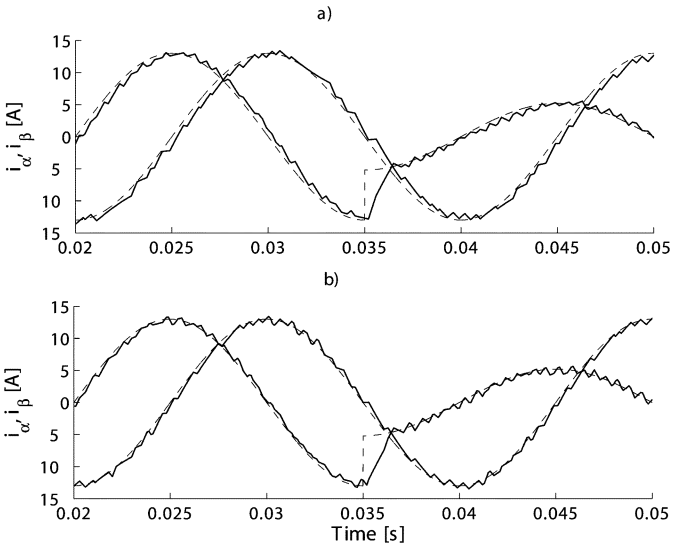


Fig. 14. Effect of model errors in the load current. a) With a -50% error in the load inductance L. b) With a $+50\%$ error in the load inductance L.

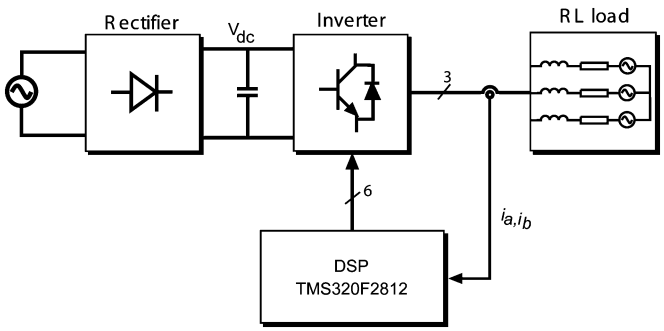


Fig. 15. Experimental system setup overview.

and $L = 12$ mH, the back-EMF has a constant amplitude of 34 V and a frequency of 50 Hz. The DC link voltage is set to $V_{dc} = 100$ V.

An overview of the system is shown in Fig. 15. The DC link capacitor is fed by a single-phase rectifier, the inverter is built with an IGBT module and connected to a active RL load. Two currents are measured to close the current control loop and the DSP is programmed to perform the algorithm and to generate the gate signals for the IGBTs.

For the implementation, a simplified version of (13) was used, neglecting the effect of the resistance R. The equation is reduced to

$$\mathbf{i}(k+1) = \mathbf{i}(k) + \frac{T_s}{L}[\mathbf{v}(k+1) - \hat{\mathbf{e}}(k)]. \tag{17}$$

The dynamic response of the system with a sampling time $T_s = 100$ μs is shown in Fig. 16 for a step change in the amplitude of i_α^* (from 4 A to 2 A at time $t = 0.025$), the reference is followed with fast dynamic behavior without affecting i_β. This result is very similar to the one presented in Fig. 11, validating the model used for simulation.

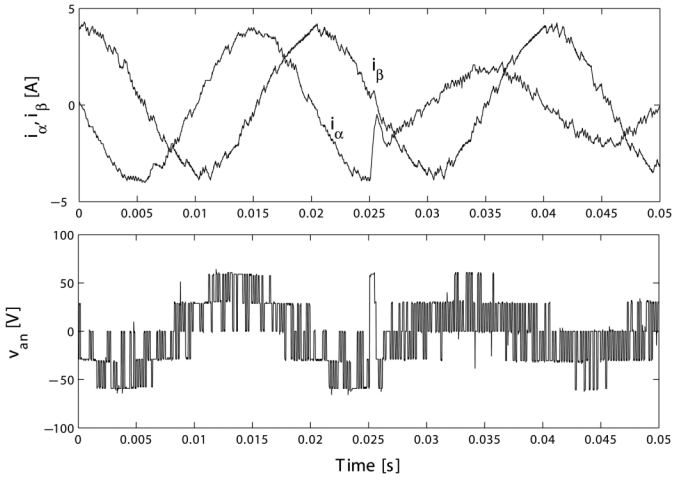


Fig. 16. Experimental results with $T_s = 100\,\mu$s for a step on i_α^*. Top: Load currents. Bottom: Load voltage.

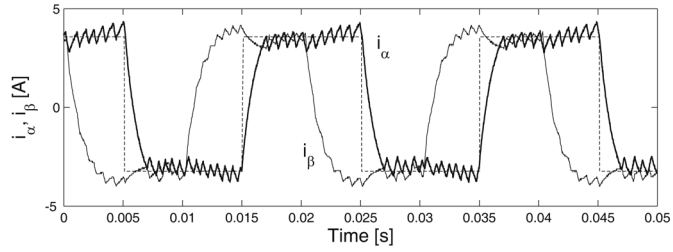


Fig. 17. Experimental results with $T_s = 100\,\mu$s for a square reference current.

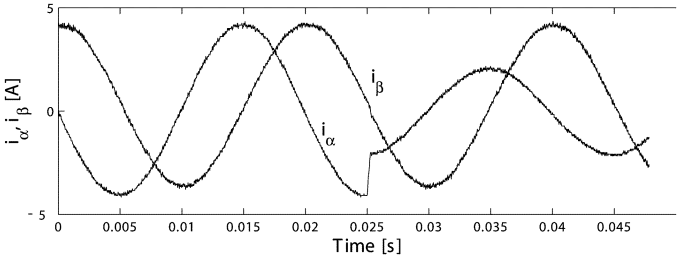


Fig. 18. Experimental results with $T_s = 20\,\mu$s. Load currents for a step on i_α^*.

Fig. 16 also shows the load voltage for the predictive current control. With this control strategy, no modulator needs to be implemented and the control signals for the switches are generated directly by the predictive controller.

The performance of the control strategy using a square waveform in orthogonal coordinates as a reference current, as shown in Fig. 17. In this test, the i_α current correctly follows the reference but some interaction appears between the currents observed in i_β current.

The effect of changing the sampling frequency was tested. The same test applied for Fig. 16 is considered using a sampling time $T_s = 20$ μs. It is shown, in Fig. 18, that using a smaller sampling time, a major separation between the fundamental and switching harmonics is obtained. The overall performance of the control is improved, achieving a very good reference tracking, and a better transient response.

The voltage spectrum obtained for $T_s = 100$ μs and $T_s = 20$ μs is shown in Fig. 19. It is observed that for $T_s = 100$ μs

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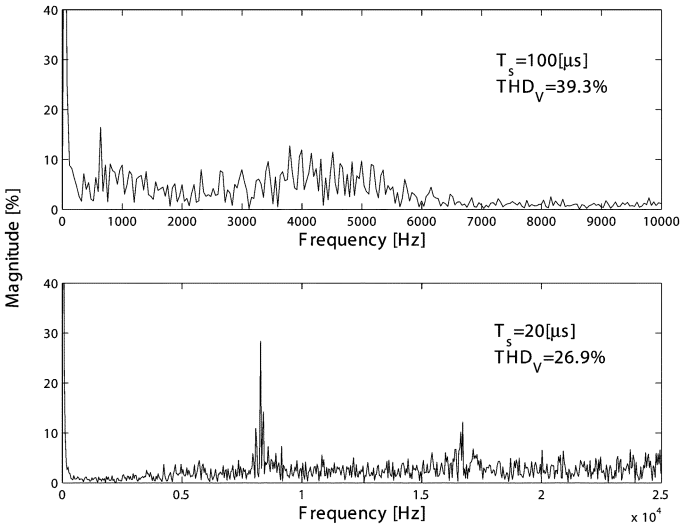


Fig. 19. Load voltage spectrum for the experimental results.

the spectrum is distributed as for the simulation results shown in Fig. 12, but concentrated near 1 KHz. For the case of using a smaller sampling time $T_s = 20\ \mu$s, the voltage spectrum is spread over a wider frequency range and the harmonic content presents a smaller amplitude appearing a clear peak near 8 KHz.

The calculation time used by the DSP to perform the current control under the conditions previously mentioned is less than 7 μs. The control algorithm is simple to implement and the remaining processing time and resources can be used for other tasks such as speed control.

VII. Comments and Conclusion

A predictive current control strategy and its practical implementation has been presented. It has been shown that the proposed method controls very effectively the load currents having a good dynamic response and compares very well with the classical methods.

The implementation of the control strategy has been discussed. The method is simple and the control algorithm is easy to implement on a DSP. The proposed strategy avoids the use of linear and nonlinear controllers. In addition, it is not necessary to include any type of modulator. The drive signals for the IGBTs are generated directly by the control.

Due to the importance of the model used for prediction, the robustness of the control method has been studied for errors in the values of load inductance and resistance of the model. The effect of the resistance can be neglected. The performance of the control deteriorates if the estimated inductance is lower than the real value, but it is almost not affected for an overestimated inductance value. This makes it preferable to overestimate a bit the inductance value.

The strategy introduced in this paper is very simple and powerful, and advantageously considers the discrete nature of power converters and microprocessors. In addition, the high calculation power of today's existing DSPs makes this method very attractive to control power converters.

These results show that predictive control is a very powerful tool with a conceptually different approach which opens up new possibilities for power converters control. The method can be applied without major changes to any type of converter and variables to be controlled.

References

[1] J. Holtz, "Pulsewidth modulation electronic power conversion," *Proc. IEEE*, vol. 82, pp. 1194–1214, Aug. 1994.

[2] M. P. Kazmierkowski, R. Krishnan, and F. Blaabjerg, *Control in Power Electronics*. New York: Academic, 2002.

[3] N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics*, 2nd ed. New York: Wiley, 1995.

[4] R. Kennel and A. Linder, "Predictive control of inverter supplied electrical drives," in *Proc. Conf. Record Power Electronics Specialists*, Galway, Ireland, Jun. 2000, pp. 761–766.

[5] R. Kennel, A. Linder, and M. Linke, "Generalized predictive control (GPC) ready for use in drive applications ?," in *Proc. Conf. Record Power Electronics Specialists*, Vancouver, Canada, Jun. 2001.

[6] H. Le-Huy, K. Slimani, and P. Viarouge, "Analisis and implementation of a real-time predictive current controller for permanent-magnet synchronous servo drives," *IEEE Trans. Ind. Electron.*, vol. 41, no. 1, pp. 110–117, Feb. 1994.

[7] O. Kukrer, "Discrete-time current control of voltage-fed three-phase PWM inverters," *IEEE Trans. Ind. Electron.*, vol. 11, no. 2, pp. 260–269, Mar. 1996.

[8] L. Malesani, P. Mattavelli, and S. Buso, "Robust dead-beat current control for PWM rectifier and active filters," *IEEE Trans. Ind. Appl.*, vol. 35, no. 3, pp. 613–620, May/Jun. 1999.

[9] P. Mattavelli, G. Spiazzi, and P. Tenti, "Predictive digital control of power factor preregulators," in *Proc. Conf. Record Power Electronics Specialists*, Mexico, 2003, pp. 1703–1708.

[10] W. Zhang, G. Feng, and Y.-F. Liu, "Analysis and implementation of a new PFC digital control method," in *Proc. Conf. Record Power Electronics Specialists*, Mexico, 2003, pp. 335–340.

[11] A. Dell'Aquila, A. Lecci, and M. Liserre, "Predictive control of half-bridge single-phase active filter," in *Proc. Record 10th Eur. Conf. Power Electron. Appl.*, Sep. 2003, CD-ROM.

[12] J. Holtz and S. Stadtfeld, "A predictive controller for the stator current vector of ac machines fed from a switched voltage source," in *Proc. Int. Power Electron. Conf.*, Tokyo, 1983, pp. 1665–1675.

[13] S. Muller, U. Ammann, and S. Rees, "New modulation strategy for a matrix converter with a very small mains filter," in *Proc. Power Electron. Specialists Conf.*, Mexico, 2003, pp. 1275–1280.

[14] J. Rodríguez, J. Pontt, C. Silva, P. Cortés, S. Rees, and U. Ammann, "Predictive direct torque control of an induction machine," in *Proc. Power Electron. Motion Control Conf.*, Riga, Latvia, Sep. 2–4, 2004, CD-ROM.

[15] J. Rodríguez, J. Pontt, P. Correa, P. Lezana, and P. Cortés, "Predictive power control of an ac/dc/ac converter," in *Proc. IEEE 40th Annual Meeting Industry Appl. Society*, Hong Kong, Oct. 2–6, 2005, pp. 934–939.

[16] J. Rodríguez, J. Pontt, P. Cortés, and R. Vargas, "Predictive control of a three-phase neutral point clamped inverter," in *Proc. Power Electron. Specialists Conf.*, Recife, Brazil, Jun. 12–16, 2005, pp. 1364–1369.



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